

Checkpoint

14.6: Directional Derivatives

Problem 1. *If we want to find the directional derivative in the direction of $\vec{v} = \langle 19, -2 \rangle$ what is wrong with $\vec{v}$?*

Problem 2. *What is one way to interpret the formula for a directional derivative in the direction of $\hat{u} = \langle u_1, u_2 \rangle$ which is $\mathcal{D}_{\hat{u}}f(x, y) = f_x u_1 + f_y u_2$?*

Problem 3. *If we package our partial in a vector known as the gradient $\langle f_x, f_y \rangle = \nabla f(x, y)$ what does $\nabla f(x, y)$ and $|\nabla f(x, y)|$ tell us?*

Problem 4. *If $f_x = x + e^y$ and $f_y = \ln(x^2 y)$ find the directional derivative in the direction that is made by an angle of $\pi/3$ from the x -axis.*

1 Answers

1. It is not a unit vector since $|\vec{v}| = \sqrt{365}$
2. A weighted average of our partials. We can think of u_1 and u_2 being the percentage we weight our partial in the x and partial in the y direction.
3. the direction of steepest ascent from any point and the slope of the steepest ascent.
4. $(x + e^y)(\frac{1}{2}) + (\ln(x^2y))(\frac{\sqrt{3}}{2})$